MANASWI CHOWDARY MAKKENA

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EDUCATION

Arizona State University **December 2026**
Bachelor of Science in Computer Science (GPA: 3.81) *Tempe, AZ*

WORK EXPERIENCE

AI Engineer **August 2026 – Present**
Mach-AI (DealPilot) *Los Angeles, CA*
- Shipped an AI sales lead summary feature passing all 25 validation snapshots by designing typed Pydantic API contracts.
- Set a 3s per story config by benchmarking Claude Haiku, Sonnet, and Opus and prompt engineering 6 story types.
- Matched an AI scorer on 23 of 25 leads with no LLM call via a deterministic model with 16 unit and integration tests.
- Kept 100% of lead results available during model failures by building a fallback flow logging every model call for review.

Software Engineer Intern **January 2026 – May 2026**
Endless Moments LLC *Austin, TX*
- Automated candidate screening with bias flagging via a document extraction pipeline indexing resumes in Elasticsearch.
- Delivered 30+ full stack HR, PTO, and benefits pages in 5 sprints by building Next.js and TypeScript UIs on REST APIs.
- Enabled real time PTO approvals via an event driven WebSockets and Kafka flow, covered by 11 automated and E2E tests.
- Hardened PostgreSQL with 28 indexes, 11 constraints, and 14 integrity tests, and fixed Docker and Kubernetes networking.
- Sped up debugging across HR, PTO, and benefits portals by building centralized error logging for every backend service.

Software Developer Intern **August 2025 – January 2026**
Neuramod *Denver, CO*
- Cut task completion steps by 40% for 200+ beta users by redesigning 4 mobile workflows and their Firebase data flows.
- Served real time operational health data in 3 releases via Firestore schemas and 6 REST APIs on Google Cloud Platform.
- Increased daily logging frequency by 25% by shipping Firebase Cloud Messaging push alerts across 3 notification types.
- Owned schema design, code reviews, and 3 production releases end to end for a health logging app with 200+ users.

PROJECTS

Cite | *Python, OpenAI API, Tavily, sentence-transformers, FastAPI, pytest, GitHub Actions*
- Caught 78 of 80 planted citation errors (98%) and accepted 20 of 20 true claims by verifying every judge quote in code.
- Raised supported sentences from 85.7% to 89.0% on the same 20 answers with local embeddings and a disk vector cache.
- Served answers in a streaming FastAPI app with rate limits and a daily spend cap, at about $0.0015 per question.

Settle | *Java 21, Spring Boot, Kafka, PostgreSQL, Docker, Testcontainers, GitHub Actions*
- Built a payments ledger on Kafka and PostgreSQL with a transactional outbox, idempotency keys, and double entry.
- Lost 0 of 20,000 payments and applied 0 twice while killing workers 12 times and the API twice, with 2,115 client retries.
- Verified the full flow against real PostgreSQL and Kafka with Testcontainers in CI, running 1 API and 3 workers in Docker.

Gauge | *Python, AWS Bedrock, Streamlit, GitHub Actions*
- Compared 8 model setups on 64 tasks; Nova Lite scored 95% against Claude Haiku's 98% at about 18 times lower cost.
- Chose rule based scoring after hand labeling 62 answers: rules agreed 98.4% of the time and the LLM judge 83.9%.
- Added a CI gate that caught a deliberately broken prompt (accuracy 95% to 61%) in 62 seconds for under half a cent.

Repo Scanner | *Python, OpenAI, GitHub REST API, FastAPI, Rich, PyPI*
- Published a PyPI CLI that summarizes any GitHub repo in a median of 5.0s, with all 10 test repos under 15s.
- Cut tokens sent to the model by 84% (median, 6 Python repos) using AST outlines, and served it via FastAPI on Render.
- Handled 8 failure cases with typed errors, shipped Rich, JSON, and Markdown output, and wrote 63 pytest tests.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, Kotlin, C++, SQL, HTML/CSS, OOP, Data Structures, Design Patterns
Frontend & Backend: Spring Boot, FastAPI, Django, Node.js, React, Next.js, REST API Design, WebSockets, Kafka
AI & LLMs: Claude API, OpenAI API, AWS Bedrock, RAG, Embeddings, AI Agents, Prompt Engineering, LLM Evaluation
Cloud & DevOps: AWS (Lambda, S3, Bedrock), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Git, Agile, Scrum
Data & Testing: PostgreSQL, Redis, Elasticsearch, Firestore, JUnit, Testcontainers, pytest, Jest, Integration and E2E Testing